

DAMR (%) within each student error type of the test split; n = contexts per error type

	Mistake identification (desired: Yes)					Mistake location (desired: Yes)					Revealing the answer (desired: No)					Providing guidance (desired: Yes)				
Numerical calculation error (n = 38)	97	39	68	63	84	97	37	61	50	71	100	95	68	84	71	95	24	21	29	26
Missing or extra quantity (n = 71)	94	35	59	58	75	92	31	49	45	58	99	93	66	87	82	85	21	13	15	24
Missing or incorrect factual knowledge (n = 42)	98	60	60	69	79	95	55	43	50	69	98	86	64	83	79	88	40	10	24	33
Misunderstanding of the question (n = 85)	95	38	62	65	87	93	38	47	52	69	98	98	75	92	82	87	31	11	22	26
None of the above (n = 26)	100	35	81	73	92	100	35	65	58	77	100	96	65	100	81	96	27	31	38	23
Reached the correct solution but proceeded further (n = 21)	90	48	62	76	95	90	43	57	67	86	100	95	76	86	86	90	24	19	19	43
Unit conversion error (n = 15)	87	53	53	67	93	80	47	40	53	73	100	93	80	87	93	87	33	7	20	20
	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%
	Actionability (desired: Yes)					Coherence (desired: Yes)					Tutor tone (desired: Encouraging)					Human-likeness (desired: Yes)				
Numerical calculation error (n = 38)	95	24	21	29	29	100	100	95	92	95	100	76	21	58	34	100	92	50	76	63
Missing or extra quantity (n = 71)	85	25	13	21	25	100	97	89	99	99	100	66	23	45	35	99	80	42	72	61
Missing or incorrect factual knowledge (n = 42)	88	40	10	33	33	100	98	88	98	93	100	74	19	48	40	100	81	48	74	60
Misunderstanding of the question (n = 85)	87	32	11	27	28	100	100	92	96	96	100	76	24	49	33	100	89	48	72	62
None of the above (n = 26)	96	27	31	38	27	100	100	88	85	96	100	65	35	62	35	100	85	62	77	69
Reached the correct solution but proceeded further (n = 21)	90	24	19	24	43	95	100	90	95	95	100	57	38	38	43	100	81	52	62	81
Unit conversion error (n = 15)	87	40	7	20	20	100	73	100	93	100	100	47	33	53	20	100	73	47	80	40
	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%	Base	SFT	DPO 10%	DPO 30%	DPO 50%